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Name _____

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SID _____

School/Department _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^\infty e^{-t} t^{\nu-1} dt, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

c	h	e	m _e	u	N _A	k _B
3.0×10 ⁸ m/s	6.63×10 ⁻³⁴ J·s 1.24×10 ³ eV·nm/c	1.6×10 ⁻¹⁹ C	9.11×10 ⁻³¹ kg 0.511 MeV/c ²	931.494 MeV/c ²	6.02×10 ²³ mol ⁻¹	8.617×10 ⁻⁵ eV/K

The quark combinations of some particles: p—uud n—udd Λ⁰—uds Σ⁺—uus

Quantum numbers of three quarks:

Quark symbols	Charge numbers	Spin	Baryon numbers	Strangeness
u	2/3	1/2	1/3	0
d	-1/3	1/2	1/3	0
s	-1/3	1/2	1/3	-1

Select five out of the following six problems.

1.(20 pts.) When neutrons are in thermal equilibrium within a reactor, they satisfy the equipartition theorem.

(a) If the temperature of the reactor is 300 K, what is the average kinetic energy $\bar{\varepsilon}$ of such a thermal neutron?

(b) The mass of neutron is known to be 1.0087 u, What is the corresponding de Broglie wavelength λ ?

Solution: (a) According to the equipartition theorem (or the law of equipartition) of classical statistical mechanics, the average kinetic energy per degree of freedom is $1/2 k_B T$. Here, the translational motion of the neutrons is in the 3 dimensional space and the degree of freedom is 3. Thus the average kinetic energy is

$$\bar{\varepsilon} = \frac{3}{2} k_B T = \frac{3}{2} \times 8.617 \times 10^{-5} \text{ eV/K} \times 300 \text{ K} = 3.88 \times 10^{-2} \text{ eV}. \quad \left(\bar{\varepsilon} = \frac{3}{2} k_B T = \frac{3}{2} \cdot \frac{1}{40} \text{ eV} = \frac{3}{80} \text{ eV} \right)$$

(b) Since $\bar{\varepsilon} \ll m_n c^2$, it is the non-relativistic case. Thus the corresponding average momentum of the neutrons is

$$\bar{p} = \sqrt{2m_n \bar{\varepsilon}} \quad \text{and}$$

$$\begin{aligned} \lambda &= \frac{h}{\bar{p}} = \frac{hc}{c\sqrt{2m_n \bar{\varepsilon}}} = \frac{hc}{1240 \text{ nm} \cdot \sqrt{2m_n c^2 \bar{\varepsilon}}} \\ &= \frac{1240 \text{ nm}}{\sqrt{2 \cdot 1.0087 \cdot 931.5 \text{ eV} \cdot 3.88 \times 10^{-2} \text{ eV}}} = 0.145 \text{ nm} \end{aligned}$$

(For the relativistic case, using the relations $E = mc^2 = m_0 c^2 + E_K$ (mass energy=rest (mass) energy+kinetic

energy) and $E^2 = m_0^2 c^4 + p^2 c^2$, we obtain

$$p = \frac{1}{c} \sqrt{E^2 - m_0^2 c^4} = \frac{1}{c} \sqrt{(E - m_0 c^2)(E + m_0 c^2)} = \frac{1}{c} \sqrt{E_K (E_K + 2m_0 c^2)}$$

and

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{E_K (E_K + 2m_0 c^2)}}.$$

For the non-relativistic case, $E_K \ll m_0 c^2$, we have

$$\lambda = \frac{hc}{\sqrt{E_K(E_K + 2m_0c^2)}} = \frac{hc}{\sqrt{2m_0c^2 E_K \left(1 + \frac{E_K}{2m_0c^2}\right)}} \approx \frac{h}{\sqrt{2mE_K}}.$$

For the extreme-relativistic case, $E = pc$, we have $\lambda = \frac{hc}{E}$.

2. (20 pts.) After long effort, in 1902, Marie and Pierre Curie succeeded in separating from uranium ore the first substantial quantity of radium, one decigram of pure RaCl_2 (radium chloride). Actually the radium they obtained was the radioactive isotope ^{226}Ra , which has a half-life of 1600 a.

(a) How many nuclei of ^{226}Ra had they isolated?

(b) What is the average life time τ of the isotope ^{226}Ra .

(c) What is the decay rate R of their sample at a late time t ?

(The atomic mass of Cl(chlorine) is 35.5u.)

Solution: (a) The number of isolated ^{226}Ra nuclei is

$$N_0 = \frac{m}{M(\text{RaCl}_2)} N_A = \frac{0.1\text{g}}{(226 + 2 \times 35.5)\text{g/mol}} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 2.0 \times 10^{20}.$$

$$(b) \quad N = N_0 e^{-\lambda t}, \quad \lambda = \frac{\ln 2}{t_{1/2}}, \quad -dN = \lambda N_0 e^{-\lambda t} dt$$

$$\begin{aligned} \tau &= \frac{\int t(-dN)}{N_0} = \int_0^\infty \lambda t e^{-\lambda t} dt \stackrel{u=\lambda t}{=} \frac{1}{\lambda} \int_0^\infty u e^{-u} du = -\frac{1}{\lambda} \int_0^\infty u d(e^{-u}) = -\frac{1}{\lambda} u e^{-u} \Big|_0^\infty + \frac{1}{\lambda} \int_0^\infty e^{-u} du \\ &= -\frac{1}{\lambda} e^{-u} \Big|_0^\infty = \frac{1}{\lambda} = \frac{t_{1/2}}{\ln 2} = 2.04 \times 10^3 \text{ a}. \end{aligned}$$

$$(c) \quad R(t) = -\frac{dN}{dt} = \lambda N_0 e^{-\lambda t} = R_0 e^{-\lambda t}, \quad \lambda = \frac{\ln 2}{t_{1/2}} = 0.433 \times 10^{-3} \text{ a}^{-1} = 1.38 \times 10^{-11} \text{ s}^{-1},$$

$$R_0 = 2.76 \times 10^9 \text{ s}^{-1} \quad (\text{Bq} \quad \text{Becquerel})$$

3.(20 pts.) An electron of a stationary hydrogen atom passes from the fifth level to the ground state, with a photon emitted in the process.

(a) Calculate the energy E_1 of the ground state and the energy E_5 of the fifth level, according to the Bohr theory.

(b) Considering the recoil of the hydrogen atom, what is the energy E_γ of the emitted photon, to the order of $(E_5 - E_1)^2 / mc^2$ where $m = 938.782 \text{ MeV}/c^2$ is the rest mass of the hydrogen atom?

(c) To the same order of magnitude, what is the recoil energy E_{recoil} of the hydrogen atom?

$$\text{Solution: (a)} \quad E_n = -13.6 \frac{1}{n^2} \text{ eV}, \quad E_1 = -13.6 \text{ eV}, \quad E_5 = -0.544 \text{ eV}.$$

(b) The conservation of momentum gives

$$p_\gamma = p_H. \quad (1)$$

The conservation of energy gives

$$\sqrt{m^2 c^4 + p_H^2 c^2} + E_\gamma = mc^2 + (E_5 - E_1), \quad E_\gamma = p_\gamma c.$$

Substituting $p_H = p_\gamma$ into above equation, we have

$$\sqrt{m^2 c^4 + E_\gamma^2} + E_\gamma = mc^2 + (E_5 - E_1) \quad \text{or} \quad \sqrt{m^2 c^4 + E_\gamma^2} = mc^2 + (E_5 - E_1) - E_\gamma.$$

To square the both sides of above equation, we have

$$m^2 c^4 + E_\gamma^2 = m^2 c^4 + (E_5 - E_1)^2 + E_\gamma^2 + 2mc^2(E_5 - E_1) - 2mc^2 E_\gamma - 2(E_5 - E_1)E_\gamma$$

or

$$2[mc^2 + (E_5 - E_1)]E_\gamma = (E_5 - E_1)^2 + 2mc^2(E_5 - E_1) - 2(E_5 - E_1)E_\gamma$$

Thus

$$E_\gamma = E_5 - E_1 - \frac{(E_5 - E_1)^2}{2[mc^2 + (E_5 - E_1)]} \approx E_5 - E_1 - \frac{(E_5 - E_1)^2}{2mc^2} \quad (\text{Noting that } E_5 - E_1 \ll mc^2)$$

(c)

$$E_{\text{recoil}} = \sqrt{m^2 c^4 + p_H^2 c^2} - mc^2 = \sqrt{m^2 c^4 + p_\gamma^2 c^2} - mc^2 = mc^2 \sqrt{1 + \frac{p_\gamma^2}{m^2 c^2}} - mc^2$$

$$\approx \frac{p_\gamma^2}{2m} = \frac{E_\gamma}{2mc^2} \approx \frac{(E_5 - E_1)^2}{2mc^2}.$$

4.(20 pts.) Black body radiation can be treated as an ideal gas of photons. As such, its density of states $D(\nu)$ in the frequency ν space is $D(\nu) = V 8\pi \nu^2 c^{-3}$ and the number of quantum states within the interval of frequency ν to $\nu + d\nu$ is $V 8\pi \nu^2 c^{-3} d\nu$, where V is the volume of the photon gas.

(a) Show that the internal energy of the photon gas is $U = \alpha T^4 V$ where T is temperature of the system and $\alpha = k_B^4 \Gamma(4) \zeta(4) / \pi^2 c^3 \hbar^3$.

(b) Find the heat capacity at constant volume C_V of the system.

(c) Find the Helmholtz free energy F and the Gibbs free energy G of the system.

(The fundamental equation of thermodynamics here can be written as $dU = TdS - pdV$.)

$$\int_0^{+\infty} \frac{x^{\nu-1}}{e^x - 1} dx = \Gamma(\nu) \zeta(\nu), \nu > 0.$$

Solution: (a)

$$U = \int_0^\infty \frac{h\nu}{e^{\beta h\nu} - 1} D(\nu) d\nu = \frac{8\pi h V}{c^3} \int_0^\infty \frac{\nu^3}{e^{\beta h\nu} - 1} d\nu \stackrel{\beta h\nu = x}{=} \frac{8\pi h V}{c^3} \frac{1}{(\beta h)^4} \int_0^\infty \frac{x^3}{e^x - 1} dx$$

$$= \frac{8\pi V k_B^4 T^4}{c^3 h^3} \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{8\pi V k_B^4 T^4}{c^3 h^3} \Gamma(4) \zeta(4) = \alpha T^4 V.$$

$$(b) \quad C_V = \left(\frac{\partial U}{\partial T} \right)_V = 4\alpha T^3 V.$$

(c) Using the relation $C_V = T \left(\frac{\partial S}{\partial T} \right)_V = 4\alpha T^3 V$, we have

$$S = \int_0^T \frac{C_V}{T} dT = S = \int_0^T \frac{4\alpha T^3 V}{T} dT = \frac{4}{3} \alpha T^3 V.$$

Therefore, the Helmholtz free energy is

$$F = U - TS = \alpha T^4 V - \frac{4}{3} \alpha T^4 V = -\frac{1}{3} \alpha T^4 V.$$

Using the relation $dF = -SdT - pdV$, we have

$$p = -\left(\frac{\partial F}{\partial V}\right)_T = -\frac{1}{3}\alpha T^3$$

Therefore, the Gibbs free energy is

$$G = U - TS + pV = F + pV = 0.$$

(Or $G = N\mu = 0$, where μ is the chemical potential of the photon gas (per particle) and is zero.)

5. (20 pts.) A one-dimensional simple harmonic oscillator with mass m and natural angular frequency ω is found to stay in the quantum state of following normalized wave function

$$\psi = c\left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2},$$

where $\alpha = (m\omega/\hbar)^{1/2}$, and c is a positive constant to be determined.

(a) Determine the constant c .

(b) Is the ψ an eigenfunction of the Hamiltonian of the oscillator?

(c) Find $\langle E \rangle$, the average energy of the oscillator.

(d) Find ΔE , the uncertainty or fluctuation of the energy of the oscillator.

Solution: (a) From

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} |\psi|^2 dx = \int_{-\infty}^{\infty} \left[c\left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \right]^2 dx \\ &= \int_{-\infty}^{\infty} \left(c^2 \frac{\alpha}{\sqrt{\pi}} e^{-\alpha^2 x^2} + 2c \frac{\alpha^2 x}{\sqrt{\pi}} e^{-\alpha^2 x^2} + \frac{\alpha^3 x^2}{\sqrt{\pi}} e^{-\alpha^2 x^2} \right) dx \\ &= \int_{-\infty}^{\infty} \left(c^2 \frac{\alpha}{\sqrt{\pi}} e^{-\alpha^2 x^2} + \frac{\alpha^3 x^2}{\sqrt{\pi}} e^{-\alpha^2 x^2} \right) dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} (c^2 e^{-u^2} + u^2 e^{-u^2}) du \\ &= \frac{c^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} du + \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{c^2}{\sqrt{\pi}} \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt + \frac{1}{\sqrt{\pi}} \int_0^{\infty} t^{\frac{1}{2}} e^{-t} dt \\ &= \frac{c^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) + \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = c^2 + \frac{1}{2}, \end{aligned}$$

we obtain $c^2 = \frac{1}{2}$ or $c = \frac{\sqrt{2}}{2}$.

$$(b) \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

$$\text{Noting that } \frac{d^2}{dx^2} \left(e^{-\frac{1}{2}\alpha^2 x^2} \right) = \frac{d}{dx} \left(-\alpha^2 x e^{-\frac{1}{2}\alpha^2 x^2} \right) = -\alpha^2 e^{-\frac{1}{2}\alpha^2 x^2} + \alpha^4 x^2 e^{-\frac{1}{2}\alpha^2 x^2},$$

and

$$\begin{aligned} \frac{d^2}{dx^2} \left(x e^{-\frac{1}{2}\alpha^2 x^2} \right) &= \frac{d}{dx} \left(e^{-\frac{1}{2}\alpha^2 x^2} - \alpha^2 x^2 e^{-\frac{1}{2}\alpha^2 x^2} \right) = -\alpha^2 x e^{-\frac{1}{2}\alpha^2 x^2} - 2\alpha^2 x e^{-\frac{1}{2}\alpha^2 x^2} + \alpha^4 x^3 e^{-\frac{1}{2}\alpha^2 x^2}, \\ &= -3\alpha^2 x e^{-\frac{1}{2}\alpha^2 x^2} + \alpha^4 x^3 e^{-\frac{1}{2}\alpha^2 x^2} \end{aligned}$$

we have

$$\begin{aligned}
\hat{H}e^{-\frac{1}{2}\alpha^2 x^2} &= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left(e^{-\frac{1}{2}\alpha^2 x^2} \right) + \frac{1}{2} m \omega^2 x^2 \left(e^{-\frac{1}{2}\alpha^2 x^2} \right) = \frac{\alpha^2 \hbar^2}{2m} e^{-\frac{1}{2}\alpha^2 x^2} - \frac{\hbar^2 \alpha^4}{2m} x^2 e^{-\frac{1}{2}\alpha^2 x^2} + \frac{1}{2} m \omega^2 x^2 e^{-\frac{1}{2}\alpha^2 x^2} \\
&= \frac{\frac{m\omega}{\hbar} \hbar^2}{2m} e^{-\frac{1}{2}\alpha^2 x^2} - \frac{\hbar^2 \left(\frac{m\omega}{\hbar} \right)^2}{2m} x^2 e^{-\frac{1}{2}\alpha^2 x^2} + \frac{1}{2} m \omega^2 x^2 e^{-\frac{1}{2}\alpha^2 x^2} = \frac{\hbar\omega}{2} e^{-\frac{1}{2}\alpha^2 x^2};
\end{aligned}$$

$$\begin{aligned}
\hat{H}x e^{-\frac{1}{2}\alpha^2 x^2} &= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left(x e^{-\frac{1}{2}\alpha^2 x^2} \right) + \frac{1}{2} m \omega^2 x^2 \left(x e^{-\frac{1}{2}\alpha^2 x^2} \right) \\
&= \frac{3\alpha^2 \hbar^2}{2m} x e^{-\frac{1}{2}\alpha^2 x^2} - \frac{\hbar^2 \alpha^4}{2m} x^3 e^{-\frac{1}{2}\alpha^2 x^2} + \frac{1}{2} m \omega^2 x^3 e^{-\frac{1}{2}\alpha^2 x^2} \\
&= \frac{3\frac{m\omega}{\hbar} \hbar^2}{2m} x e^{-\frac{1}{2}\alpha^2 x^2} - \frac{\hbar^2 \left(\frac{m\omega}{\hbar} \right)^2}{2m} x^3 e^{-\frac{1}{2}\alpha^2 x^2} + \frac{1}{2} m \omega^2 x^3 e^{-\frac{1}{2}\alpha^2 x^2} = \frac{3\hbar\omega}{2} x e^{-\frac{1}{2}\alpha^2 x^2}.
\end{aligned}$$

Therefore

$$\hat{H}\psi = \frac{\hbar\omega}{2} \left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \frac{3\hbar\omega}{2} \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \neq E\psi,$$

ψ is not an eigenfunction of $\hat{H}$.

(c)

$$\begin{aligned}
\langle E \rangle &= \int_{-\infty}^{\infty} \psi^* \hat{H} \psi dx \\
&= \int_{-\infty}^{\infty} \left[\left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \right] \left[\frac{\hbar\omega}{2} \left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \frac{3\hbar\omega}{2} \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \right] dx \\
&= \frac{\hbar\omega}{2} \int_{-\infty}^{\infty} \frac{\alpha}{2\sqrt{\pi}} e^{-\alpha^2 x^2} dx + 2\hbar\omega \int_{-\infty}^{\infty} \frac{\alpha^2 x}{\sqrt{2\pi}} e^{-\alpha^2 x^2} dx + \frac{3\hbar\omega}{2} \int_{-\infty}^{\infty} \frac{\alpha^3 x^2}{\sqrt{\pi}} e^{-\alpha^2 x^2} dx = \frac{\hbar\omega}{2} \cdot \frac{1}{2} + \frac{3\hbar\omega}{2} \cdot \frac{1}{2} = \hbar\omega.
\end{aligned}$$

(d)

$$\hat{H}^2 \psi = \hat{H}(\hat{H}\psi) = \left(\frac{\hbar\omega}{2} \right)^2 \left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{3\hbar\omega}{2} \right)^2 \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2},$$

$$\begin{aligned}
\langle E^2 \rangle &= \int_{-\infty}^{\infty} \psi^* \hat{H}^2 \psi dx \\
&= \int_{-\infty}^{\infty} \left[\left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \right] \left[\left(\frac{\hbar\omega}{2} \right)^2 \left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} + \left(\frac{3\hbar\omega}{2} \right)^2 \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2 x^2} \right] dx \\
&= \left(\frac{\hbar\omega}{2} \right)^2 \int_{-\infty}^{\infty} \frac{\alpha}{2\sqrt{\pi}} e^{-\alpha^2 x^2} dx + \left(\frac{3\hbar\omega}{2} \right)^2 \int_{-\infty}^{\infty} \frac{\alpha^3 x^2}{\sqrt{\pi}} e^{-\alpha^2 x^2} dx = \frac{5}{4} (\hbar\omega)^2.
\end{aligned}$$

$$\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \frac{\hbar\omega}{2}.$$

6. (20 pts.) For each of the following reactions state whether or not it is forbidden. If it is forbidden, state a conservation law that is violated.

- (a) $\Lambda^0 \rightarrow p + \pi^0$
- (b) $\bar{p} + p \rightarrow \mu^+ + e^-$
- (c) $n \rightarrow p + e^- + \nu_e$
- (d) $n + \nu_e \rightarrow p + e^-$
- (e) $\Sigma^+ \rightarrow \Lambda^0 + \pi^+$

Answer:

- (a) $\Lambda^0 \rightarrow p + \pi^0$ Q
- (b) $\bar{p} + p \rightarrow \mu^+ + e^-$ L_e and L_μ
- (c) $n \rightarrow p + e^- + \nu_e$ L_e
- (d) $n + \nu_e \rightarrow p + e^-$ not forbidden weak interaction (neutrino)
- (e) $\Sigma^+ \rightarrow \Lambda^0 + \pi^+$ not forbidden strong interaction
 $uus \rightarrow uds + u\bar{d}$ (creation of $d \bar{d}$ pair)